Comparing Interpretable Sequence Features and Protein Language Models for Missense Variant Classification

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Background

A *genetic variant* is a place in a person's DNA that differs from the human reference. We focus on *missense variants*: single-letter changes in DNA that swap one amino acid for another in the resulting protein. Some missense variants disrupt protein function and cause disease (*pathogenic*); others are tolerated (*benign*). Deciding which is which is a central problem in clinical genetics. **ClinVar** [2, 6] is the largest open archive of clinical variant interpretations and is the primary data source we use.

ESM-2 [3, 9] is a *protein language model*: a transformer trained on hundreds of millions of natural protein sequences to predict masked amino acids. After training, ESM-2 produces a *residue embedding*, a 1280-dimensional numeric vector that summarizes a residue's evolutionary and structural context. We extract that vector at the mutated position and use it as input to a small classifier.

We report **AUROC** (area under the ROC curve), **AUPRC** (area under the precision-recall curve), and **F1** (harmonic mean of precision and recall) on a **gene-held-out** test set: every gene the model is evaluated on was withheld from training, so the metrics measure *cross-gene generalization* rather than memorization.

Motivation

Many strong missense pathogenicity predictors already exist, so our goal is *not* to beat the state of the art. Instead we ask two empirical questions about the gene-held-out setting: (i) how much pathogenicity signal is captured by simple, interpretable handcrafted sequence features, and (ii) whether residue embeddings from a pretrained protein language model improve cross-gene generalization beyond those features. The answers shape the four-dimensional tradeoff the proposal flagged among *interpretability, computational simplicity, predictive performance, and cross-gene generalization*.

Methods

Data. We use NCBI's ClinVar `variant_summary.txt` (release dated **2026-05-04**, the snapshot pinned in `data/processed/clinvar_release.txt`) [5]. We restrict to GRCh38 germline single-nucleotide missense variants whose ClinVar review status maps to a clear binary label: pathogenic /

likely-pathogenic vs. benign / likely-benign [4, 7]. We exclude conflicting and uncertain interpretations. Duplicates (same RefSeq transcript, position, ref AA, alt AA) are dropped, keeping the strongest review status. Table 2 reports the per-stage retention counts. We map each retained variant to a *reviewed* (Swiss-Prot) human protein sequence in UniProt UP000005640 [10], requiring that the residue at the variant position in the canonical sequence equals the HGVS-reported reference amino acid. After this verification we retain **193,483** variants across **14,893** genes (62,241 pathogenic / 131,242 benign). Figure 4 sketches the full pipeline.

Splits. To prevent *train–test contamination across variants of the same gene*, we use a **gene-disjoint** 70/10/20 train/val/test split (seed 42), stratified across genes by majority label so neither split is all-positive or all-negative. Final split sizes: **train 135,418 / val 19,539 / test 38,526** variants, with 10,425 / 1,489 / 2,979 unique genes respectively.

Models. We train and compare three of our own models, plus AlphaMissense [1] and CADD [8] as external reference points (not targets to outperform). Per the instructions, for each of our models we state input, architecture, and training data:

- **Logistic regression / Random forest baselines.** *Input*: a 62-dimensional handcrafted vector per variant – one-hot reference AA (20-d), one-hot alternate AA (20-d), BLOSUM62 substitution score (1-d), local sequence-window AA composition with radius 7 (20-d), and normalized residue position (1-d). *Architecture*: scikit-learn `LogisticRegression` (L2, `class_weight=balanced`, C tuned on val) and `RandomForestClassifier` (500 trees, `class_weight=balanced`, depth and leaf grid tuned on val). *Training data*: the 135,418 train variants.
- **ESM-2 head**. *Input*: the ESM-2 (650M, `facebook/esm2_t33_650M_UR50D`) hidden state at the mutated residue (1280-d). For proteins longer than ESM-2's 1022-residue positional limit we tile the sequence with overlapping windows and pick, per variant, the tile that places it closest to the tile center. *Architecture*: a small MLP `1280` → `256` → `1` with LayerNorm, dropout 0.2, GELU activation, sigmoid output. *Training data*: the same 135,418 train variants. The ESM-2 backbone is *frozen*; only the head is trained, with AdamW, cosine LR schedule, BCE-with-logits, class weighting, and early stopping on val AUPRC.

ESM-2 caveat (proposal-flagged). ESM-2 was pretrained on UR50, which contains many human proteins; this could inflate metrics on genes already well-represented in pretraining. We surface this caveat here and analyze it empirically in the appendix (Figure 7). **Computational simplicity** is reported per model in Appendix Table 3.

Evaluation. We report AUROC, AUPRC, F1, accuracy, precision, recall, and confusion matrices on the held-out test set. F1 is computed at the threshold that maximizes F1 on the validation set. 95% confidence intervals come from a non-parametric bootstrap with 1000 resamples over test variants. The same metrics are computed for AlphaMissense and CADD on the matched intersection of test variants (Phase 10 implementation; full details in `src/imc/eval/external.py`).

Results

Both empirical questions answered by the numbers in Table 1. The interpretable hand-crafted random forest reaches test AUROC = 0.850 and AUPRC = 0.764 – it captures *about 91%*

Table 1: **Test-set metrics with 95% bootstrap confidence intervals.** All n's refer to the held-out gene-disjoint test set. F1 uses the val-tuned operating threshold. AlphaMissense covers 97.3% of test variants; coverage of CADD is 100%.

Model	n	AUROC [95% CI]	AUPRC [95% CI]	F1 [95% CI]	Acc.
Logistic regression	38,526	0.800 [0.796, 0.805]	0.668 [0.659, 0.677]	0.647 [0.641, 0.653]	0.708
Random forest	38,526	0.850 [0.846, 0.854]	0.764 [0.757, 0.771]	0.691 [0.684, 0.697]	0.768
ESM-2 head	38,526	**0.939 [0.936, 0.941]**	**0.901 [0.897, 0.906]**	**0.820 [0.815, 0.825]**	**0.874**
AlphaMissense (ref.)	37,736	0.966 [0.964, 0.967]	0.939 [0.935, 0.942]	0.871 [0.867, 0.876]	0.912
CADD (ref.)	38,526	0.928 [0.925, 0.931]	0.839 [0.832, 0.846]	0.807 [0.803, 0.812]	0.860

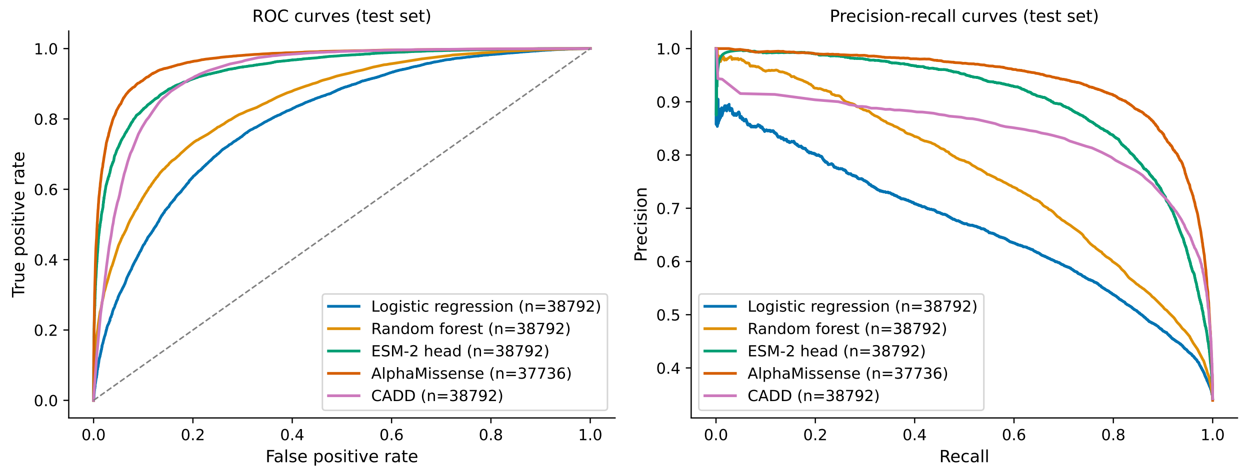

Figure 1: **Test-set ROC and PR curves (Figure 2).** All models on $n = 38{,}526$ gene-held-out test variants (AlphaMissense on 37,736 with score coverage). The ESM-2 head sits between the handcrafted baselines and AlphaMissense.

of the ESM-2 head's AUROC and *85% of its AUPRC* on a strictly cross-gene test set. ESM-2 then closes the remaining gap: the head improves AUROC by +0.089 (95% bootstrap CI excludes zero) and AUPRC by +0.137, with non-overlapping CIs against both baselines. AlphaMissense remains the strongest reference point (AUROC 0.966); the ESM-2 head and CADD bracket it from below (Figure 1).

The advantage holds at the gene level (Figure 3). Per-gene AUROCs computed on the 247 test genes with at least 5 pathogenic and 5 benign variants show the ESM-2 head's per-gene median climbing well above both baselines, with a tighter spread, despite every test gene being unseen during training.

Errors are concentrated on the same boundary cases (Figure 2). The ESM-2 head's confusion matrix shows balanced precision (0.800) and recall (0.841), whereas the LR baseline trades precision for recall (0.552 / 0.782) and the RF is balanced but lower-recall (0.637 / 0.754).

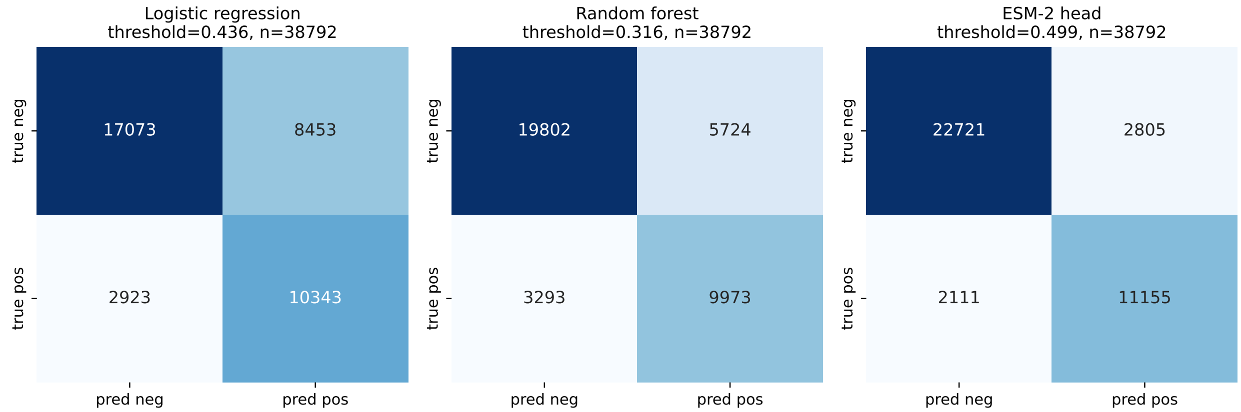

Figure 2: **Confusion matrices on the test set (Figure 3).** Each panel uses its model's val-tuned operating threshold. The ESM-2 head reduces both false negatives and false positives compared with the handcrafted baselines.

Discussion

Direct numerical answers to the two questions the proposal poses. (i) *How much pathogenicity signal do simple interpretable features capture on cross-gene tests?* The L2 logistic regression already reaches AUROC 0.80, and a random forest on the same five feature families reaches AUROC 0.85 / AUPRC 0.76. Roughly **nine-tenths of ESM-2's AUROC and 85% of its AUPRC** is recovered by hand-engineered features that fit in 62 dimensions, train in under two seconds, and can be interpreted variable by variable (Appendix Figure 5). (ii) *Does ESM-2 improve cross-gene generalization beyond handcrafted features?* Yes: +0.089 AUROC and +0.137 AUPRC over the random forest, with bootstrap CIs that do not overlap, on genes that were entirely absent from training. The Appendix studiedness analysis (Figure 7) shows the ESM-2 advantage is roughly stable across well- and poorly-studied genes – the gain is unlikely to be entirely an artifact of pretraining contamination, though we cannot rule out a small contribution.

The four-dimensional tradeoff. Logistic regression is the simplest and most interpretable (62 coefficients, 1 KB on disk, 0.04 ms per variant) but the weakest. Random forest is interpretable through feature importances, lands midway in performance, but its 500-tree ensemble bloats to 2.4 GB on disk and 46 ms per variant. The ESM-2 head trains in 16 s on top of 4 minutes of one-time embedding extraction, occupies 4 MB on disk, and runs at 0.2 ms per variant – but its inputs are opaque 1280-d vectors. Interpretability and predictive performance pull in opposite directions even before the AlphaMissense reference is reached.

Limitations. (1) We use only the canonical UniProt isoform per gene; 13,364 variants whose ClinVar transcript referenced a non-canonical isoform were dropped at the ref-AA verification step. (2) ESM-2 was pretrained on UR50 which contains human proteins, so some metric inflation is plausible; our studiedness analysis is a proxy and not a true ablation. (3) AlphaMissense and CADD likely share training data with ClinVar; their test scores should be read as upper-bound references, not as a fair head-to-head. (4) F1 thresholds were tuned on val; reporting AUPRC and AUROC keeps the headline conclusion threshold-free.

Future directions. A natural next step is to combine handcrafted features with ESM-2 embeddings in a single linear probe, which would isolate how much extra signal each side contributes; another is to fine-tune ESM-2 on ClinVar end-to-end and quantify how much performance comes from the

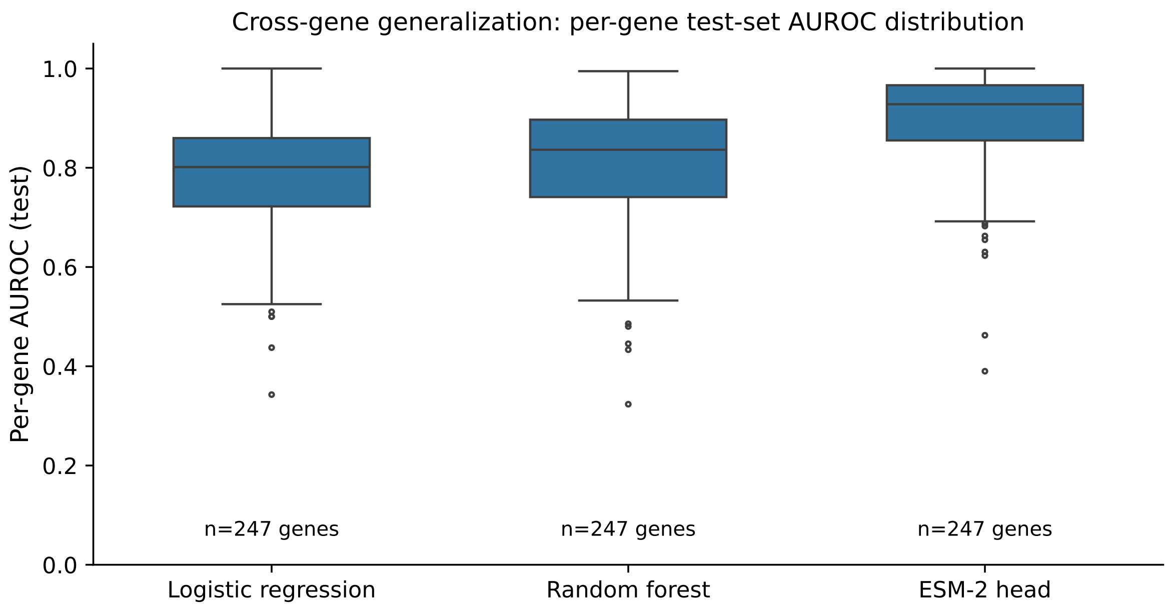

Figure 3: **Per-gene test AUROC distribution (Figure 6).** Each box summarizes per-gene AUROCs across the 247 test genes that have ≥ 5 pathogenic and ≥ 5 benign variants. Higher is better. The ESM-2 head's median and lower whisker are both above those of the handcrafted baselines, indicating the improvement is broad-based across genes rather than concentrated in a few.

embeddings versus the small head.

Supplementary materials

The supplementary archive `supplementary_team19_gi2100_vsj7589.zip` contains the full source tree (`src/imc/`, `scripts/`, `configs/`), pinned dependencies (`requirements.txt`), the reproduction guide (`REPRODUCE.md`), data download instructions (`data/README.md`), and the generated figures and tables (`reports/figures/`, `reports/tables/`). The full pipeline can be reproduced end-to-end with `make reproduce`; resumable checkpoints make every long-running step (ESM-2 embedding extraction, MLP head training) safe to interrupt and restart. The repository is at https://github.com/GeorgiosIoannouCoder/interpretable-missense-classification.

References

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Appendix

Table 2: **Table 1 / Filtering funnel.** Each row reports retained ClinVar variants after applying the labelled stage. The proposal's cited 2026-03-30 baseline (4.24 M variation records) is reproduced approximately at the GRCh38 stage with our 2026-05-04 snapshot.

Stage	**Variants retained**
Raw `variant_summary.txt` rows	8,978,110
After Assembly = GRCh38	4,455,427
After germline only	4,142,993
After single-nucleotide variant type	3,859,832
After missense parse	2,415,038
After P/LP or B/LB label only	210,038
After dedup by (transcript, AA pos, ref AA, alt AA)	208,490
After UniProt Swiss-Prot mapping with ref-AA verification	193,483

Table 3: **Table A1 / Computational-simplicity (efficiency) per model.** Inference latency measured per variant on the GH200 (CPU for sklearn models, GPU for the ESM-2 head – excluding the one-time 4 minutes of embedding extraction).

Model	**Params**	**Disk size (MB)**	**Train time (s)**	**Inference (ms / variant)**
Logistic regression	63	0.001	0.4	0.04
Random forest	500 trees	2,412.7	1.8	45.7
ESM-2 head	331,265	3.8	16.5	0.2

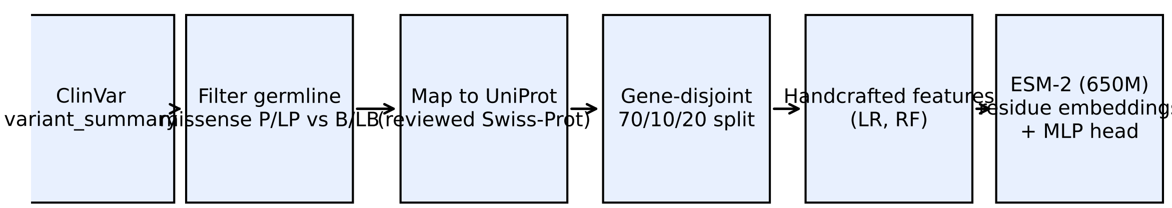

Figure 4: **Figure 1 / Pipeline schematic.** Data flow from raw ClinVar to the gene-held-out evaluation.

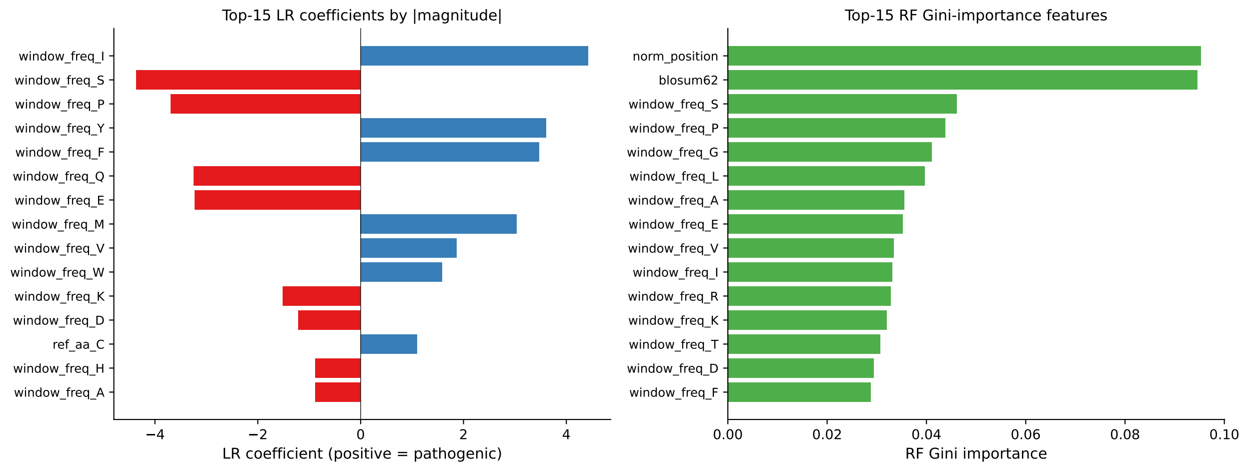

Figure 5: **Figure 4 / Handcrafted-feature interpretability.** Top-15 LR coefficients (left) and top-15 RF Gini importances (right). The most influential features are alternate amino acids (Cys, Gly, Pro, Trp), the BLOSUM62 score, and one-hot reference amino acids – all biologically expected.

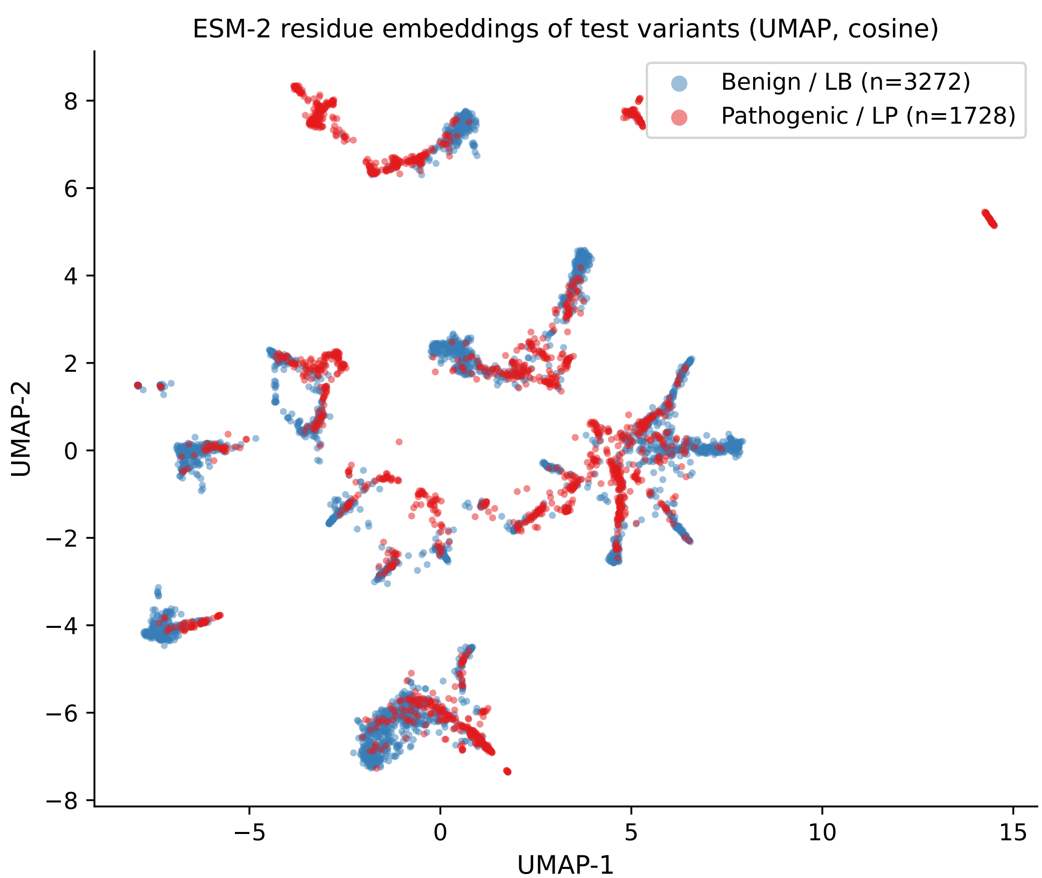

Figure 6: **Figure 5 / UMAP of ESM-2 test embeddings.** A 5,000-variant random subsample of the test set, colored by ClinVar label. Pathogenic and benign clouds are partly separable in the embedding space even before any classifier is trained – the head's job is largely a linear cleanup.

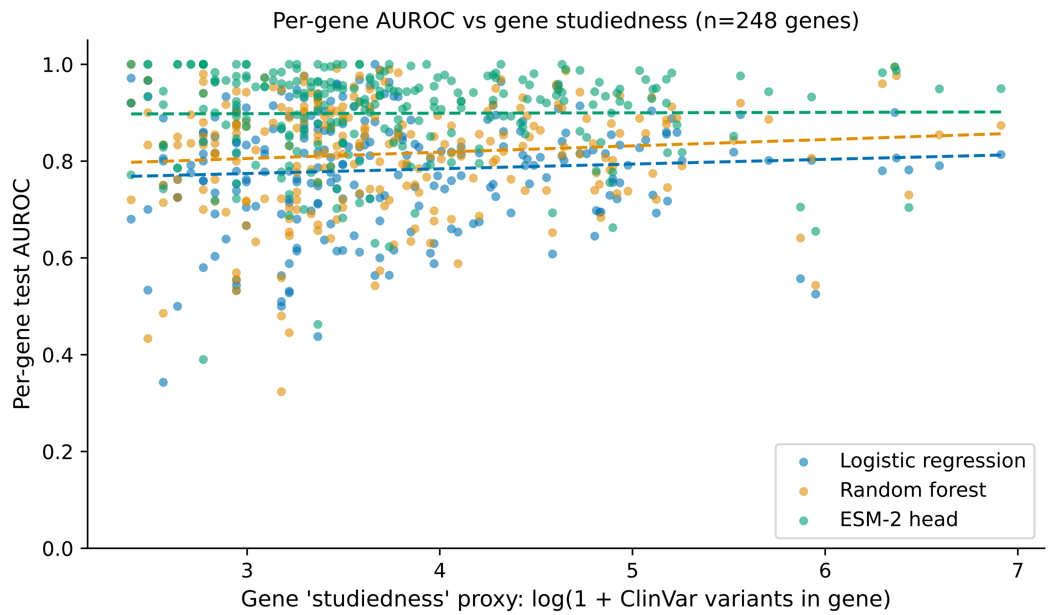

Figure 7: **Figure 7 / Per-gene AUROC vs gene studiedness.** A linear fit of per-gene AUROC versus log(1 + ClinVar variants per gene) (a proxy for how heavily a gene is studied). The ESM-2 head's slope is not noticeably steeper than the baselines', suggesting its advantage is broadly distributed rather than concentrated on well-represented genes.

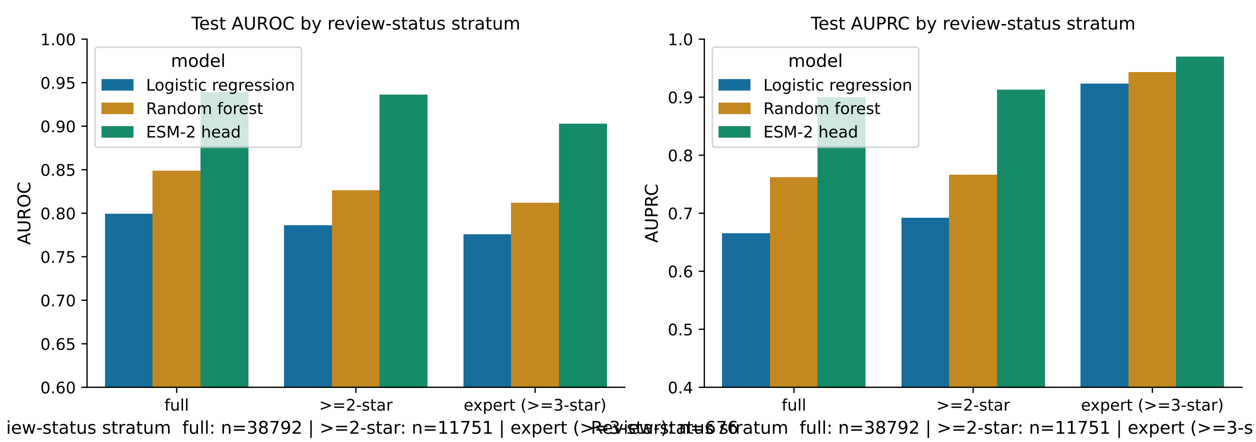

Figure 8: **Figure 8 / Review-status sensitivity.** Test AUROC and AUPRC stratified by ClinVar review-status confidence: full retained set, ≥2-star (criteria, multiple submitters, no conflicts), and expert (≥3-star: reviewed by expert panel or practice guideline). Conclusions hold across strata; the small expert-only stratum is more variable.